

King Abdullah University of Science and Technology (KAUST)
M.S. Technology Innovation and Entrepreneurship (TIE)

Research and Innovation Proposal
AI Mobility Venture-Readiness Toolkit for Saudi Smart-City Contexts

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Abstract

Saudi Arabia's smart-city agenda creates demand for tools that connect technical performance with trust, adoption, and commercialization. AI mobility concepts can predict demand, optimize services, or support planning, but many early prototypes fail before pilot deployment because teams do not test data-governance risk, user value, technical feasibility, and business logic early enough. This project proposes a venture-readiness toolkit for trustworthy AI mobility solutions in Saudi smart-city and cognitive-city contexts. The toolkit will help early-stage teams decide whether a mobility AI concept should move forward, be redesigned, or stop before costly field deployment. The project will use design science and lean-startup validation. It will combine desk research, regulatory mapping, synthetic mobility scenario modelling, risk scoring, MVP specification, and commercialization planning. Outputs will include an adoption-barrier map, a governance-to-market scoring rubric, a first-generation MVP specification, a synthetic validation scenario, and a pitch-ready commercialization roadmap. The project fits KAUST TIE because the program combines product development, MVP Studio, market analysis, prototype iteration, and Global Technology Hub exposure. The applicant will bring finance, data analysis, mobility modelling, and investment-scoring experience to this environment, then use TIE to convert that background into venture-building capability.

Keywords: KAUST TIE, technology innovation, AI mobility, smart cities, AIoT, edge computing, MVP, venture readiness, commercialization, data governance.

1. Project Fit for KAUST TIE

The project starts from a practical question: how can a team know whether an AI mobility idea deserves investment, pilot testing, or redesign? TIE fits this question because it trains students to move from ideas toward prototypes, MVPs, validation, and commercialization. The proposal therefore treats governance as part of product design, not as a separate policy layer.

KAUST TIE provides the execution setting: a 16-month structure, core courses on IoT systems, edge computing, MVP development, innovation launch, and early-stage startup development, plus a Global Technology Hub period. The project will use that structure to build a narrow but usable first-generation toolkit.

Table 1. KAUST TIE evidence base and project implication.

Program element	Relevant evidence	Project implication
TIE program model	16-month program focused on venture formation and technology innovation [1], [2].	The project must produce usable artifacts, not only a research essay.
Curriculum logic	Idea generation, market analysis, proof of concept, lab-field testing, prototype iteration, scaling, and commercialization [2].	The proposal follows an idea-to-MVP-to-roadmap sequence.
Course fit	TIE 201, TIE 203, TIE 204, TIE 211, and TIE 212 cover edge/IoT systems, systems design, MVP Studio, and startup launch [3].	The toolkit can develop across technical, design, and business work packages.
KAUST resources	Prototyping and Product Development Core Lab, Entrepreneurship Center, and Innovation Ventures [4]-[6].	The MVP can be refined through prototype support, venture mentoring, and commercialization feedback.
Faculty fit	Amr Abdel-Hamid: AIoT, edge computing, smart-city applications, rapid prototyping [9]. Mikkel Draebye: entrepreneurship and innovation funnels [8].	The project has a technical route and a venture route inside TIE.

2. Problem and Opportunity

The problem is not the absence of AI mobility ideas. The problem is the weak bridge between technical prototype quality and adoption readiness. A model may predict traffic demand or recommend route optimization, yet still fail if the data flows are unclear, the

buyer cannot see the value, deployment costs are hidden, or the pilot lacks a credible business case.

This gap matters in Saudi smart-city contexts because Vision 2030, SDAIA, and projects such as NEOM have increased demand for reliable, investable, and trusted urban technologies. Early-stage teams need a compact tool that brings governance, mobility value, technical feasibility, and commercialization into one decision process.

The proposed project addresses that gap by building a venture-readiness toolkit. It will not claim to operate a full transport platform. It will assess whether a proposed AI mobility concept has enough evidence to progress toward MVP testing and pilot design.

3. Aim, Research Questions, and Objectives

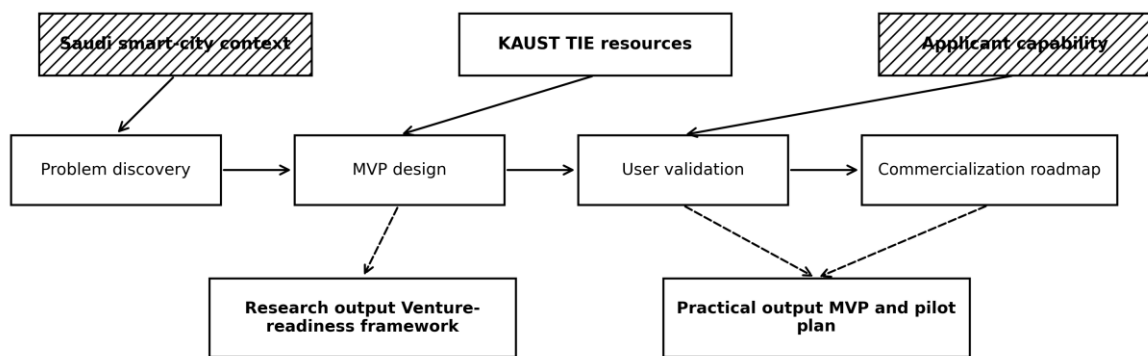
Aim: To develop and test a venture-readiness toolkit that helps transform trustworthy AI mobility concepts into market-oriented MVPs for Saudi smart-city contexts.

Table 2. Objective-question-method-deliverable alignment.

Objective	Research question	Method and data source	Deliverable
Identify adoption barriers for AI mobility solutions.	What adoption barriers affect AI mobility solutions in Saudi smart-city contexts?	Desk research; policy review; coding of barriers from official and academic sources.	Barrier map and problem-opportunity matrix.
Design a venture-readiness framework.	How can AI governance, mobility value, feasibility, and commercialization be linked in one framework?	Design-science method; synthesis of AI risk, PDPL logic, and TIE product-development stages.	First-generation venture-readiness framework.
Apply the framework to one synthetic MVP scenario.	How does the framework evaluate a synthetic AI mobility MVP scenario?	Synthetic dataset; Python scenario model; risk scoring; baseline comparison.	Scenario validation report and MVP specification.
Define a TIE-suitable commercialization roadmap.	What roadmap can move the MVP from concept to pilot readiness within TIE?	Lean-startup logic; stakeholder mapping; market assumptions; KAUST resource mapping.	Commercialization roadmap and pitch structure.

4. Conceptual Framework

The venture-readiness logic of the proposed project is summarized in Fig. 1. The framework converts three inputs into four project actions and two outputs. Saudi context defines the problem pressure. KAUST TIE resources define the execution path. The applicant's capability defines the starting skill base. The central sequence then moves from problem discovery to prototype design, validation, and commercialization planning.



Patterned boxes = contextual inputs; plain boxes = project actions or outputs; dashed arrows = validation feedback.

Figure 1. Conceptual framework linking the Saudi smart-city context, KAUST TIE resources, and applicant capability to the proposed venture-readiness project. Patterned boxes indicate contextual inputs; plain boxes indicate project actions or outputs. Solid arrows show the project sequence and contextual inputs. Dashed arrows show validation feedback into the research and practical outputs.

The framework keeps the project narrow. It does not promise a complete mobility platform. It builds an assessment layer that can sit before pilot deployment and help a team decide whether the concept has enough evidence to continue.

5. Literature and Market Rationale

The literature supports the project in four areas. AI governance clarifies risk and trust requirements. AIoT and edge systems define the technical deployment environment. Entrepreneurship literature and the TIE curriculum define the product-development path. Saudi innovation policy defines the local relevance.

Table 3. Synthesis of evidence and research gap.

Cluster	Source base	Supported point	Gap addressed
Trustworthy AI governance	NIST AI RMF; OECD AI Principles; SDAIA/PDPL [10], [12], [13].	AI systems require lifecycle risk management, transparency, privacy, and accountability.	Governance is rarely translated into early MVP go/no-go decisions.
AIoT and edge mobility systems	TIE courses; Amr Abdel-Hamid profile [3], [8].	Mobility AI can use sensors, edge computing, secure architecture, and embedded systems.	Technical feasibility alone does not prove adoption readiness.
Technology commercialization	TIE curriculum; Entrepreneurship Center; Innovation Ventures [2], [5], [6].	Venture formation requires user validation, market logic, prototype iteration, and go-to-market design.	Commercial criteria often enter after technical design rather than during early concept screening.
Saudi innovation context	Vision 2030; SDAIA; KAUST TIE launch [9]-[11].	Saudi Arabia seeks innovation, data capability, and private-sector growth.	Generic smart-city models do not address Saudi-specific trust, regulation, and venture context.
Proposed project	Design science + lean startup + synthetic scenario testing.	The project connects trust, technical feasibility, mobility value, and commercialization.	A compact toolkit will bridge AI mobility concept and MVP readiness.

6. Methodology and Methods

The study will use a design science methodology, supported by lean-startup validation. Design science is appropriate because the project will create and evaluate an artifact: a venture-readiness toolkit, a scoring rubric, and an MVP roadmap. This fits the design-science tradition in information systems, where research generates knowledge by building and evaluating artifacts such as models, methods, or instantiations. Lean-startup validation is also appropriate because the toolkit must test user value, adoption risk, and market assumptions before a team invests in full product development. The lean-startup model centers on MVP development and build-measure-learn feedback loops.

The methods will include document analysis, regulatory mapping, synthetic mobility scenario modelling, readiness scoring, business model analysis, and MVP specification. The document analysis will review official KAUST TIE material, Saudi data-governance sources, AI governance frameworks, and smart-city mobility literature. The synthetic mobility scenario will be built in Python and will test how the toolkit scores a hypothetical AI mobility concept across four dimensions: mobility value, trust and data governance,

technical feasibility, and commercialization readiness. A spreadsheet or lightweight dashboard will host the first scoring rubric. Zotero will manage references and source traceability.

The project will not use personal mobility traces. If expert interviews become possible, they will require ethics approval before recruitment. If interviews are not feasible within the 16-month timeline, the project will remain valid through document-based analysis, synthetic scenario testing, and mentor review inside the KAUST TIE environment. This keeps the methodology feasible while still producing a concrete product-development artifact.

The validation strategy will use three tests. First, the framework will be checked against official policy and AI governance sources to assess conceptual consistency. Second, it will be applied to one synthetic AI mobility MVP scenario to test whether the scoring logic can produce a clear go, revise, or stop recommendation. Third, it will be compared with a baseline assessment that considers only technical performance. If the broader framework does not outperform the technical-only baseline, the result will still clarify which governance or commercialization dimensions add little value, duplicate existing evaluation logic, or create excessive complexity.

7. Proposed MVP Concept

The applied concept is a Trustworthy AI Mobility Venture-Readiness Toolkit. The toolkit will help early-stage teams assess whether an AI mobility concept is ready to move from technical prototype to MVP. The first version can run as a structured spreadsheet, Python notebook, or lightweight dashboard before any full software build.

The toolkit will not operate traffic systems or optimize live mobility flows. It will assess pre-pilot readiness. Its role is to help teams decide whether an AI mobility concept should proceed to pilot testing, be redesigned, or be stopped before expensive deployment. This product logic fits KAUST TIE because the program includes courses and activities related

to IoT systems, edge computing, systems design, MVP Studio, launching innovation, and early-stage startup development.

Table 4. MVP components and KAUST TIE relevance.

MVP component	Function	TIE relevance
Mobility value module	Defines target user, mobility pain point, and measurable value proposition.	Customer discovery and innovation launch.
Trust and data module	Maps data type, privacy risk, explainability, PDPL relevance, and audit trail needs.	Trustworthy AI and Saudi data-governance fit.
Technical feasibility module	Assesses sensor needs, edge/AIoT constraints, deployment assumptions, and model risk.	TIE 201, TIE 203, TIE 204, and PCL resources.
Commercialization module	Builds stakeholder map, cost logic, revenue assumptions, and go-to-market path.	TIE 211, TIE 212, Entrepreneurship Center, Innovation Ventures.
Readiness score	Combines the four layers into a transparent go/revise/stop recommendation.	Final project artifact and pitch evidence.

8. Venture Potential and Product Roadmap

The first users would be smart-city labs, mobility startups, infrastructure investors, university incubators, and municipal innovation offices. These users do not need another full traffic-management platform at the pre-pilot stage. They need a sharper way to decide whether an AI mobility concept deserves pilot testing, redesign, or rejection before large resources are committed.

The product hypothesis is narrow and testable: early-stage smart-city teams lack a low-cost tool for screening AI mobility concepts before deployment. The proposed toolkit will test this hypothesis by comparing two evaluation paths. The first path will assess only technical performance. The second path will add governance, data-readiness, implementation feasibility, and market logic. If the second path produces clearer pilot decisions, the toolkit will have a stronger case as a venture-ready product.

The roadmap follows the 16-month TIE structure: problem validation, framework design, MVP specification, prototype build, validation, and Global Technology Hub refinement. This sequence fits KAUST TIE because the program moves students through market

analysis, design thinking, proof-of-concept development, lab-field testing, prototype iteration, pilot systems, scaling, and commercialization.

Table 5. Product roadmap across the 16-month TIE structure.

Stage	Timeframe	Main activity	Tangible output
Problem validation	Months 1-2	Map users, adoption barriers, and pilot-readiness problems.	Problem-opportunity brief and target-user profile.
Framework design	Months 3-5	Build scoring architecture across mobility value, trust, feasibility, and commercialization.	Venture-readiness framework and scoring rubric.
MVP specification	Months 6-7	Define dashboard screens, inputs, outputs, and report format.	MVP wireframe and functional specification.
Prototype build	Months 8-10	Create spreadsheet or Python prototype using a synthetic mobility scenario.	Working prototype or clickable MVP mock-up.
Validation and iteration	Months 11-13	Compare the toolkit with a baseline technical-only assessment; collect mentor feedback when possible.	Validation memo and revised MVP logic.
Global Hub refinement	Months 14-16	Use ecosystem exposure to refine market assumptions and pitch structure.	Pitch deck, commercialization roadmap, and final report.

9. KAUST Fit and Scholarship Rationale

The project fits KAUST TIE because it treats technology innovation as a full product-development cycle: selecting a real problem, designing a system, specifying an MVP, testing the concept with users, and building a commercialization path. This structure matches TIE's emphasis on project-based learning, prototyping cycles, market analysis, and the Global Technology Hub experience, where students refine products in an entrepreneurial environment.

The KAUST Fellowship would give this work the time, resources, and environment needed to produce tangible outputs: a tool specification, a first-generation prototype path, a validation memo, and a pitch-ready roadmap. My contribution would come from financial modelling, cost analysis, KPI dashboards, mobility and spatial analysis, and investment-scoring logic. The training plan is focused: edge AI and IoT prototyping, design science,

customer discovery, and venture validation. I am not trying to become a transport engineer in 16 months. I am trying to turn my existing finance and data-analysis background into product-building capability for smart-city decision support.

10. Expected Deliverables

Table 6. Research, product, and dissemination deliverables.

Code	Deliverable	Type	Target timeframe
D1	AI mobility adoption-barrier map.	Research deliverable	Months 1-2
D2	Venture-readiness framework integrating trust, feasibility, mobility value, and commercialization.	Research and product deliverable	Months 3-6
D3	Synthetic scenario dataset and Python notebook.	Technical deliverable	Months 5-7
D4	MVP specification: user flow, data flow, risk controls, and deployment assumptions.	Product deliverable	Months 7-9
D5	Business model canvas, stakeholder map, and go-to-market assumptions.	Entrepreneurial deliverable	Months 9-12
D6	Validation memo and revised scoring rubric.	Evidence deliverable	Months 11-13
D7	Pitch deck, commercialization roadmap, and final TIE project report.	Final deliverable	Months 13-16

11. Boundaries and Risks

The project has a narrow execution boundary. It will not deploy a live mobility system, process personal mobility traces, or claim direct adoption by NEOM, SDAIA, or a Saudi municipality. Its output will be a first-generation assessment toolkit, supported by a synthetic scenario, a validation memo, and an MVP roadmap. This boundary keeps the project feasible within the 16-month TIE structure while still producing a concrete product-development artifact.

The main risk is scope expansion. Smart-city mobility can easily expand into transport engineering, infrastructure finance, carbon accounting, geoeconomic risk, or public-sector governance. This project will use those topics only when they improve the pre-pilot

readiness score. They will not become separate research tracks. The central deliverable will remain a product-oriented toolkit that helps users decide whether an AI mobility concept should proceed to pilot testing, be redesigned, or be stopped.

A second risk is limited access to external stakeholders. If interviews or expert feedback cannot be secured, the project will remain viable through desk research, synthetic scenario testing, and mentor review within the KAUST TIE environment. This is consistent with TIE's project-based model, which moves students through proof-of-concept development, lab-field testing, prototype iteration, pilot-system thinking, scaling, and commercialization.

A third risk is weak market validation. A technically coherent toolkit may still fail to show user demand. The project will address this risk by separating technical validation from venture validation. Technical validation will test whether the framework produces clearer readiness decisions than a technical-only baseline. Venture validation will test whether the target users understand the output, see value in the assessment report, and can imagine using it before pilot deployment.

If the toolkit does not outperform the baseline or if stakeholder validation remains weak, the project will still produce useful evidence. A negative result would clarify which readiness dimensions add value and which ones create unnecessary complexity. The final output would then shift from a commercialization roadmap to a stricter research conclusion on the limits of pre-pilot assessment tools for AI mobility ventures.

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